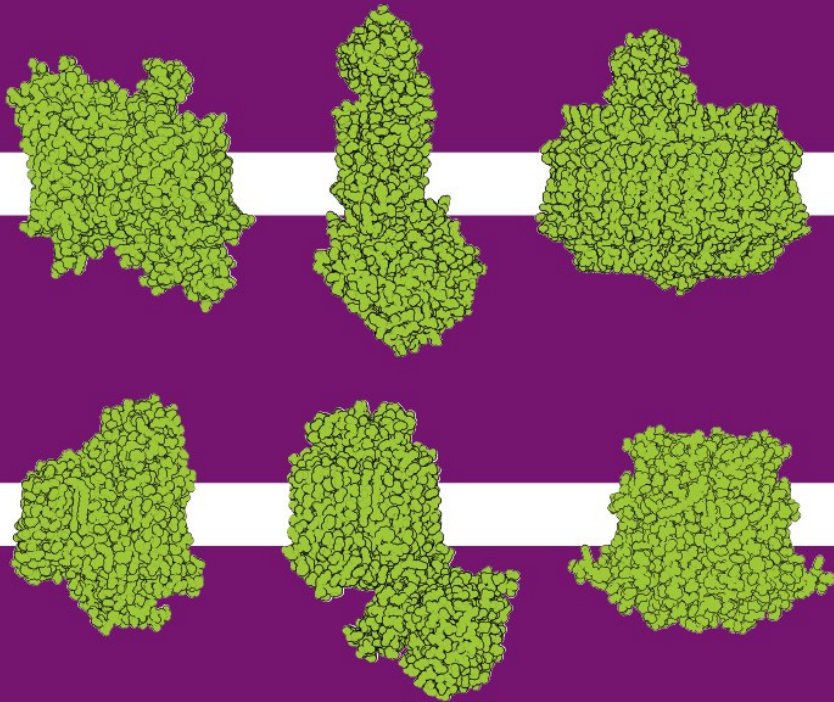


MOLECULAR BIOLOGY OF
THE CELL
SEVENTH EDITION



ALBERTS HEALD JOHNSON MORGAN RAFF ROBERTS WALTER

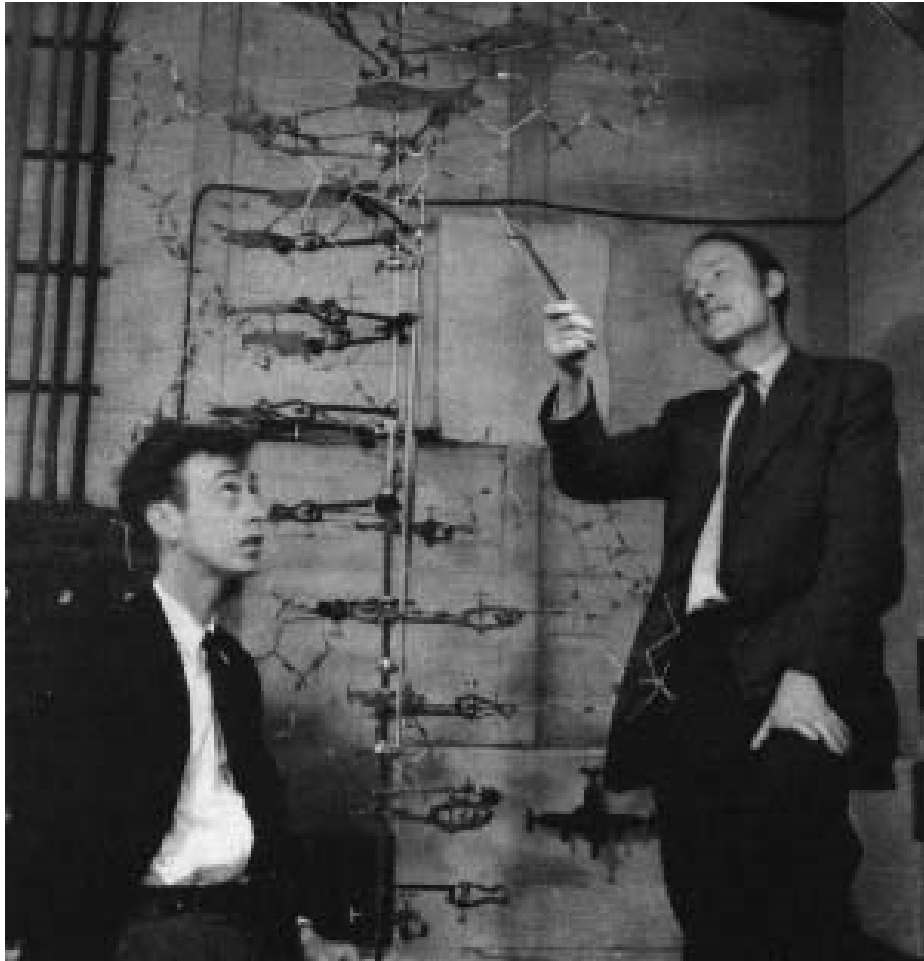
Chapter 4

DNA, Chromosomes, and Genomes

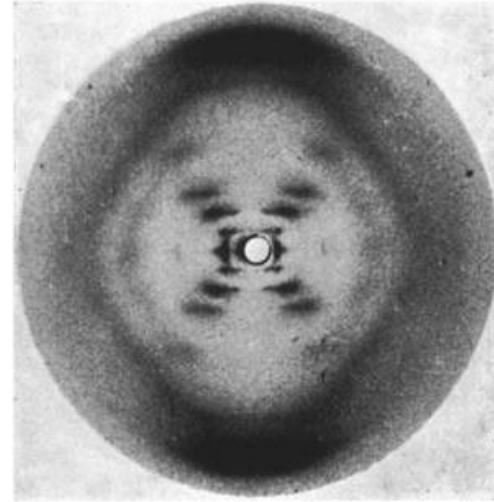
Topics to be discussed

- Structure of DNA
- Organisation of our genome
- Packaging of DNA into nucleosomes, chromatin
- Histone code
- How does chromatin function in regulating gene expression
- ENCODE project; what is it and what can we learn from it.
- Chromatin modifiers in cognition

1953 James Watson and Francis Crick proposed the double-stranded, helical, complementary, anti-parallel model for DNA.



James Watson 1928 -
Francis Crick 1916 - 2004



X-ray of DNA

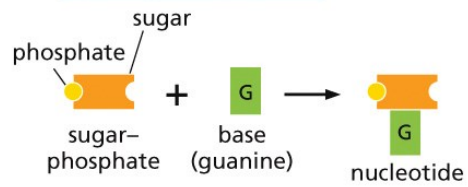


Rosalind Franklin 1920-1958

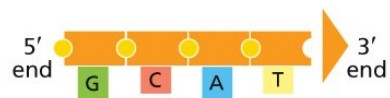
THE STRUCTURE AND FUNCTION OF DNA

- A DNA Molecule Consists of Two Complementary Chains of Nucleotides
- The Structure of DNA Provides a Mechanism for Heredity

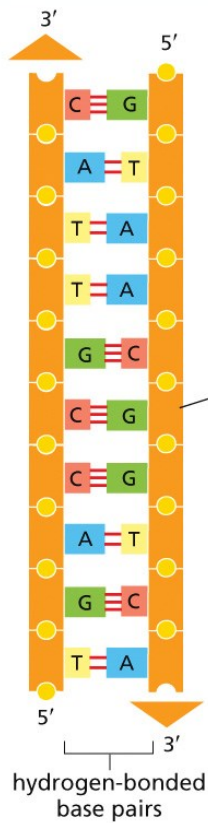
(A) building blocks of DNA



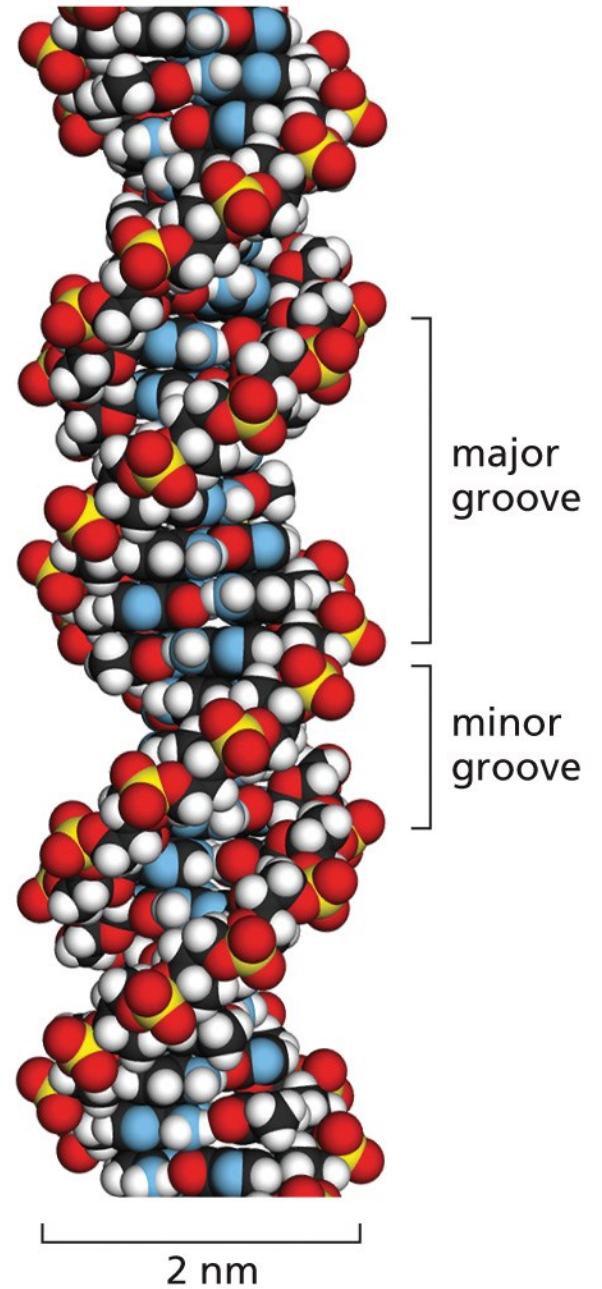
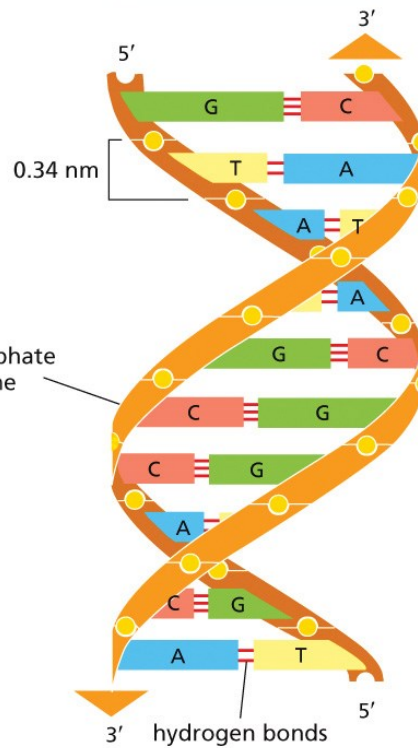
(B) DNA strand

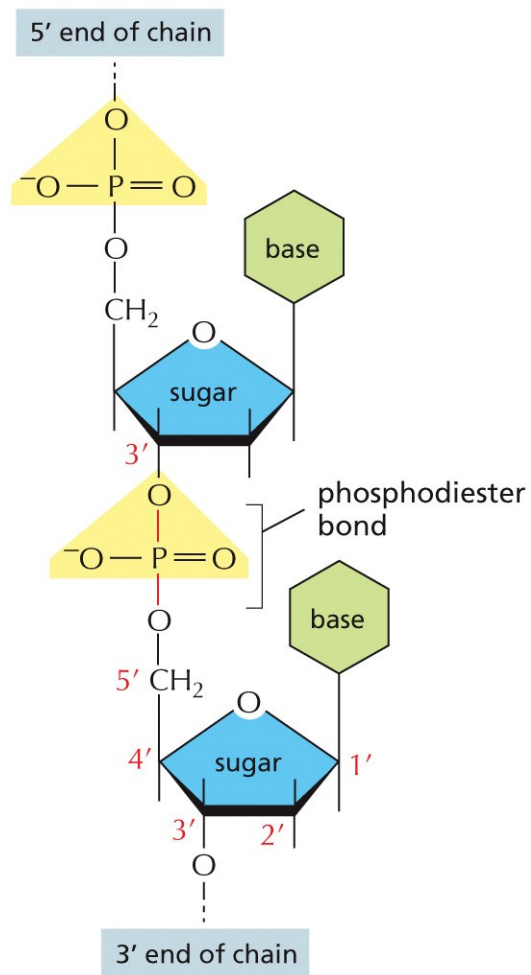


(C) double-stranded DNA

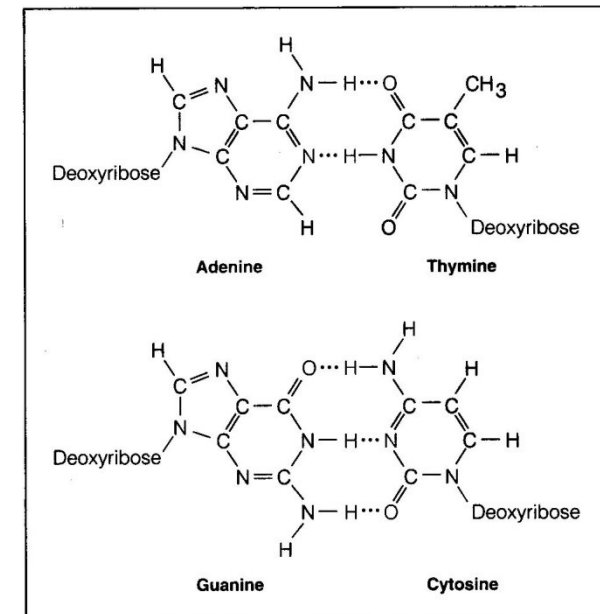
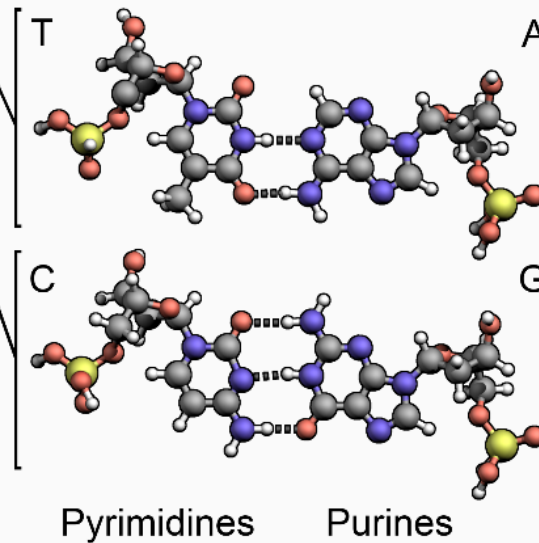
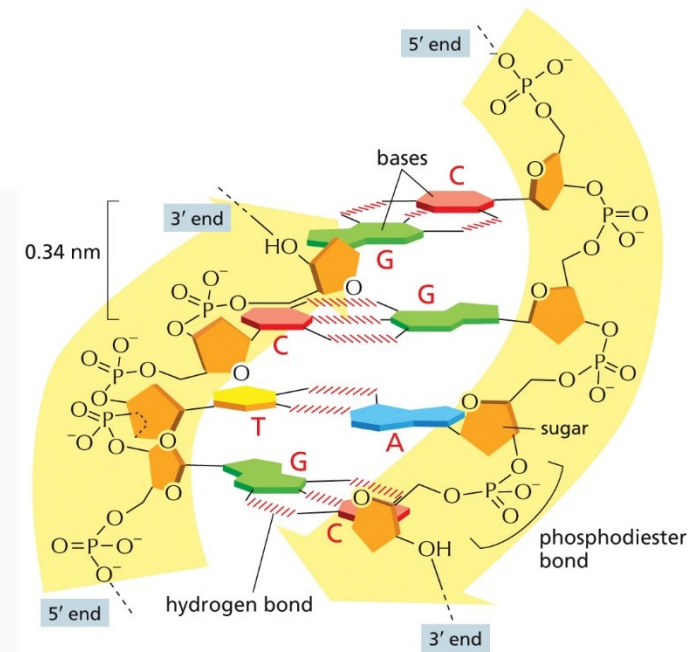
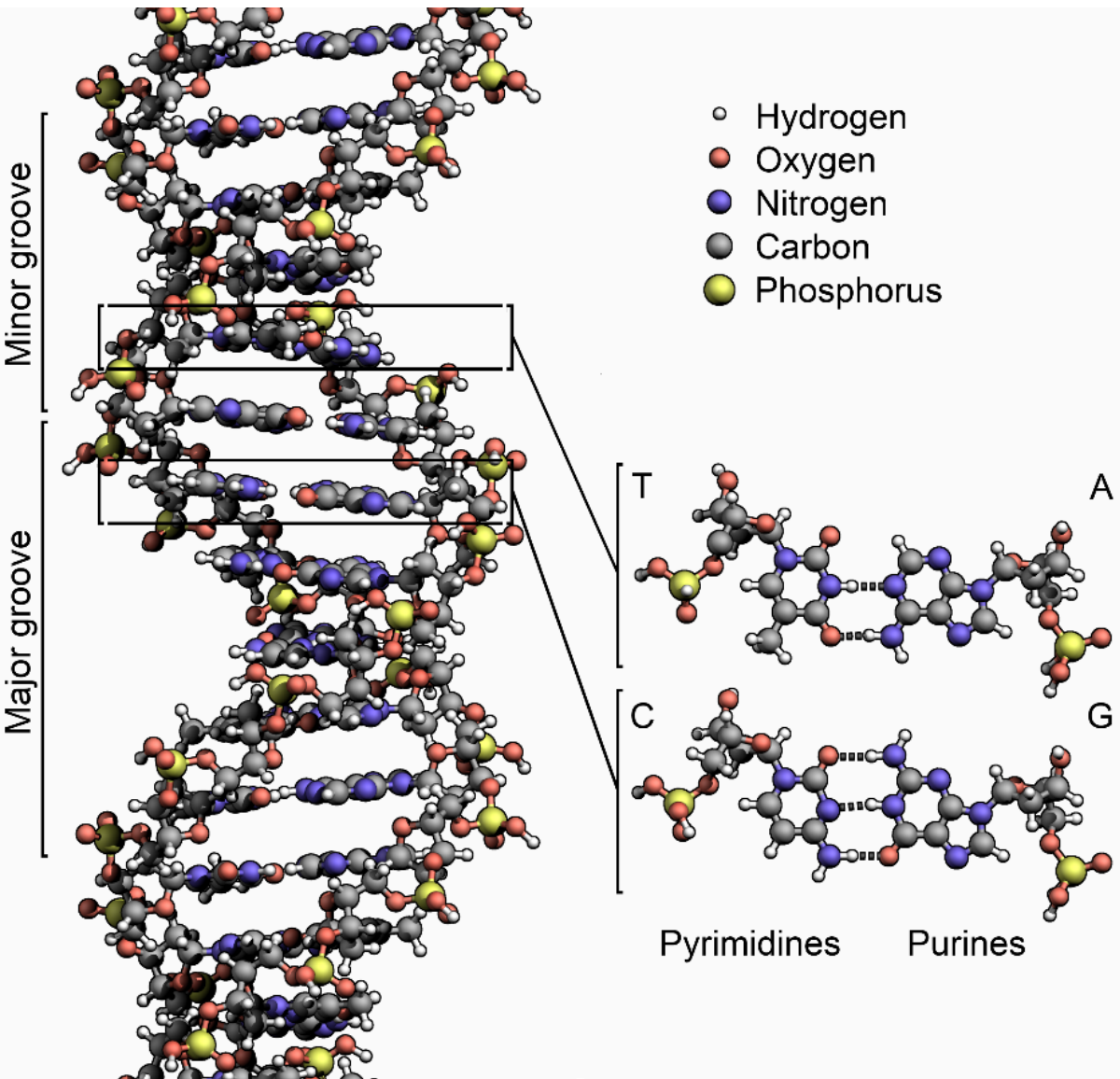


(D) DNA double helix



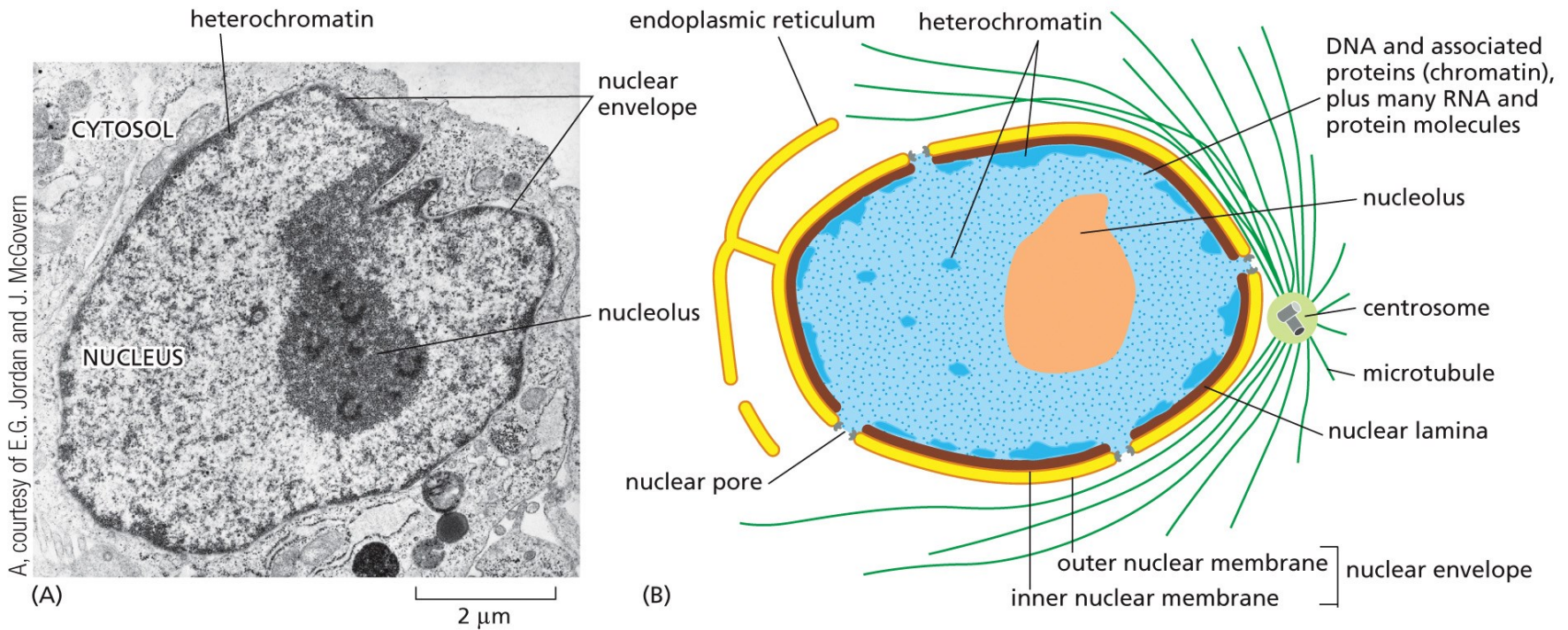


Structure of DNA

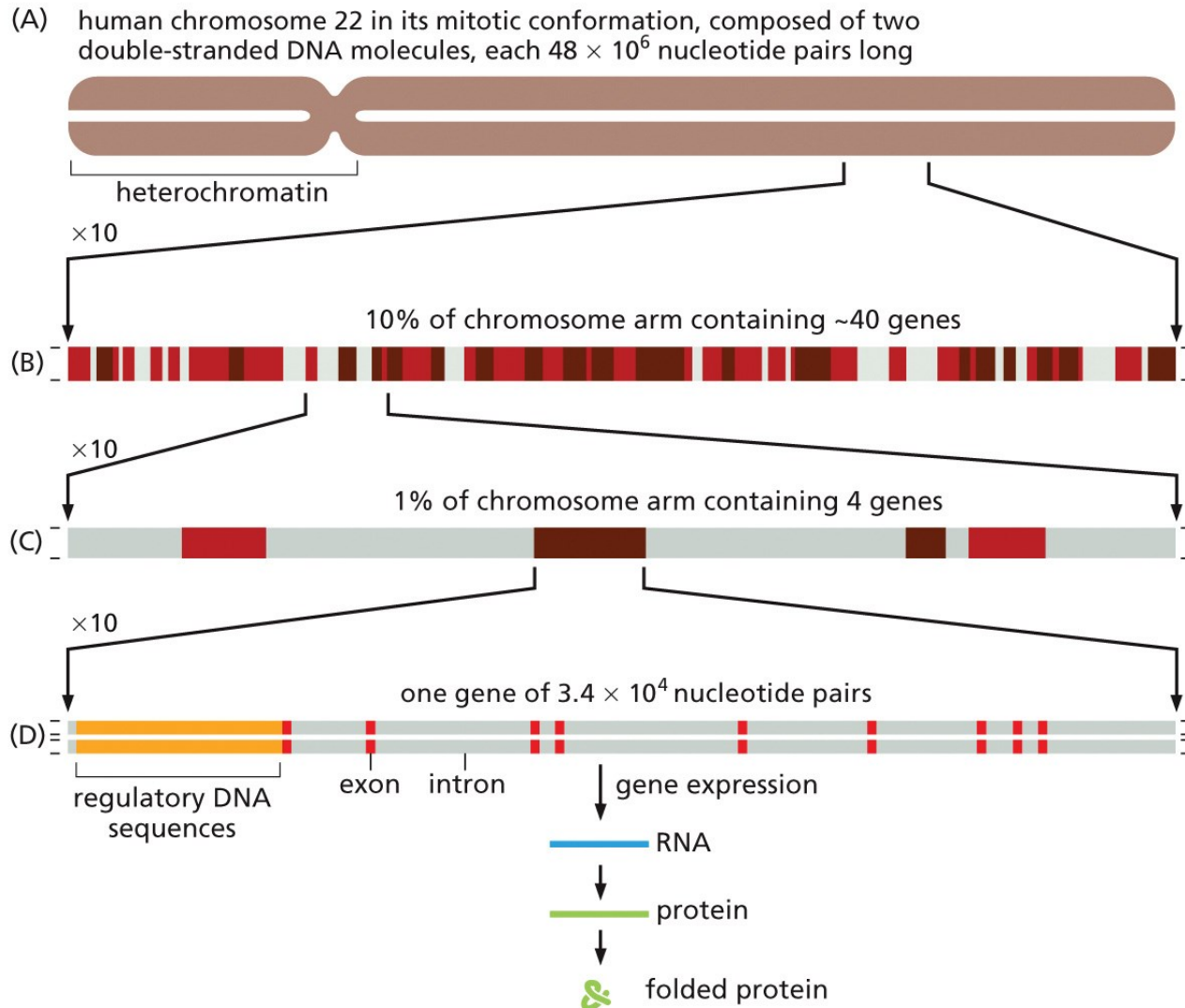


Chapter 4

- In Eucaryotes, DNA Is Enclosed in a Cell Nucleus



- Eucaryotic DNA Is Packaged into a Set of Chromosomes
- Chromosomes Contain Long Strings of Genes

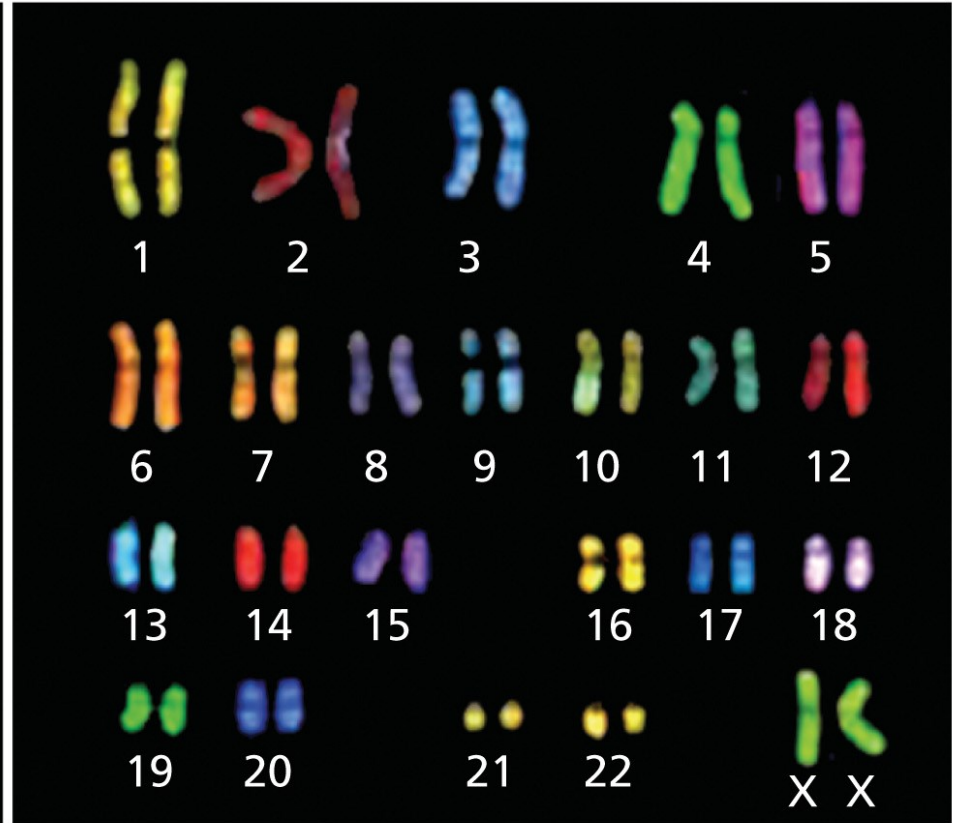


Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001.

Human Karyotype as shown by chromosome painting



(A)



(B)

10 μ m

Adapted from N. McNeil and T. Ried, *Expert Rev. Mol. Med.* 2:1-14, 2000. With permission from Cambridge University Press.

TABLE 4–1 Some Vital Statistics for the Human Genome

Protein-coding gene features	
Median size of protein produced	430 amino acids
Smallest number of exons per gene	1 (1068 of these unspliced genes)
Largest number of exons per gene	363
Median number of exons per gene	9.0
Median exon size	131 nucleotide pairs
Median intron size	1747 nucleotide pairs
Number of noncoding RNA genes	About 5000, with 4849 annotated***
Number of pseudogenes	More than 20,000****
Percentage of protein-coding DNA sequence (in exons)	1%
Percentage of DNA in other highly conserved sequences	3.5%
Percentage of DNA transcribed (in protein-coding plus annotated noncoding RNA genes)	45%
Percentage of DNA in high-copy-number repetitive elements	Approximately 50%

*The sequence of 2.85 billion nucleotides is known precisely (error rate of only about 1 in 100,000 nucleotides). The remaining DNA primarily consists of short sequences that are tandemly repeated many times over, with repeat numbers differing from one individual to the next. These highly repetitive blocks are difficult to sequence accurately.

**RefSeq genes, with at least one reviewed/validated mRNA (A. Piovesan et al., *BMC Res. Notes* 12:315–319, 2019). For genes encoding tiny proteins, see T.F. Martinez et al., *Nat. Chem. Biol.* 16:458–468, 2020).

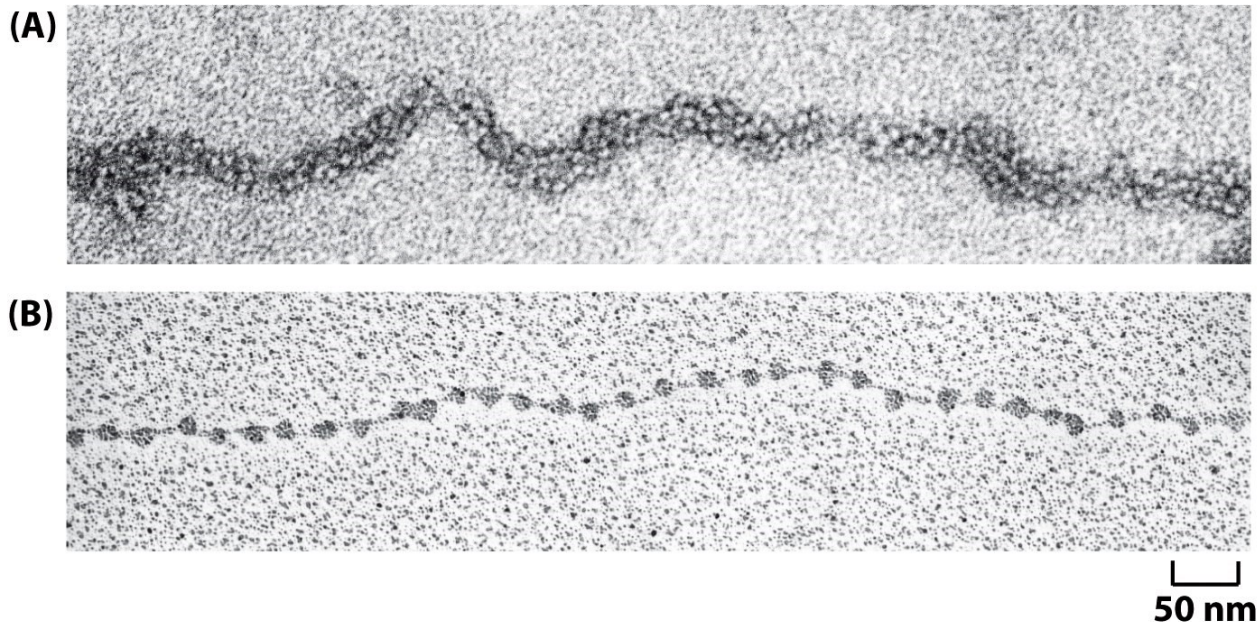
***A considerable number of potential ncRNA genes are still uncharacterized and not annotated.

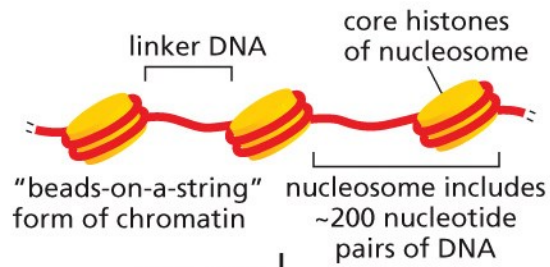
****A pseudogene is a DNA sequence closely resembling that of a functional gene but containing numerous mutations that prevent its proper expression or function; most are still uncharacterized and not annotated in databases.

Courtesy of Allison Piovesan, University of Bologna, Bologna, Italy.

CHROMOSOMAL DNA AND ITS PACKAGING IN THE CHROMATIN FIBER

- DNA Molecules Are Highly Condensed in (Mitotic) Chromosomes
- Nucleosomes Are a Basic Unit of Eucaryotic Chromosome Structure





NUCLEASE DIGESTS LINKER DNA



released nucleosome core particle

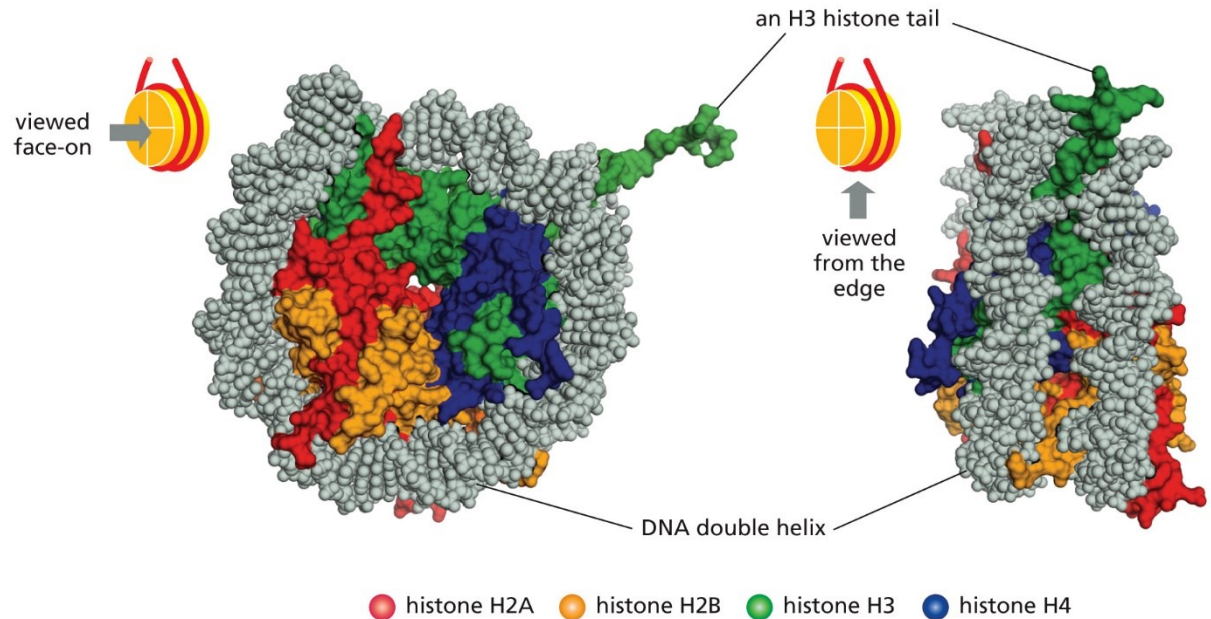
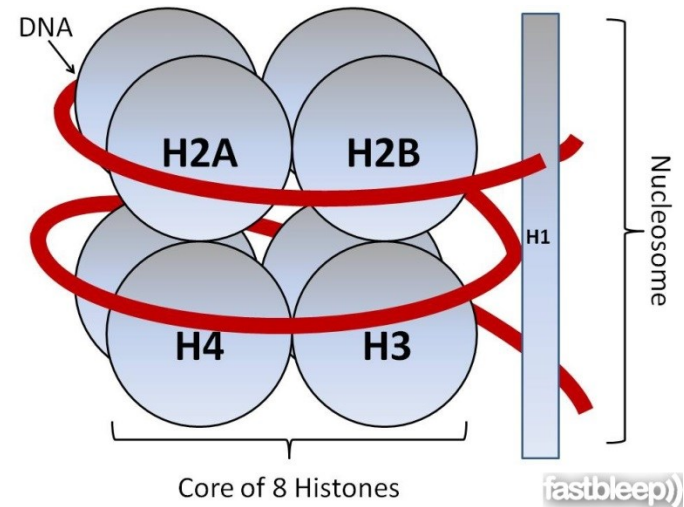
11 nm

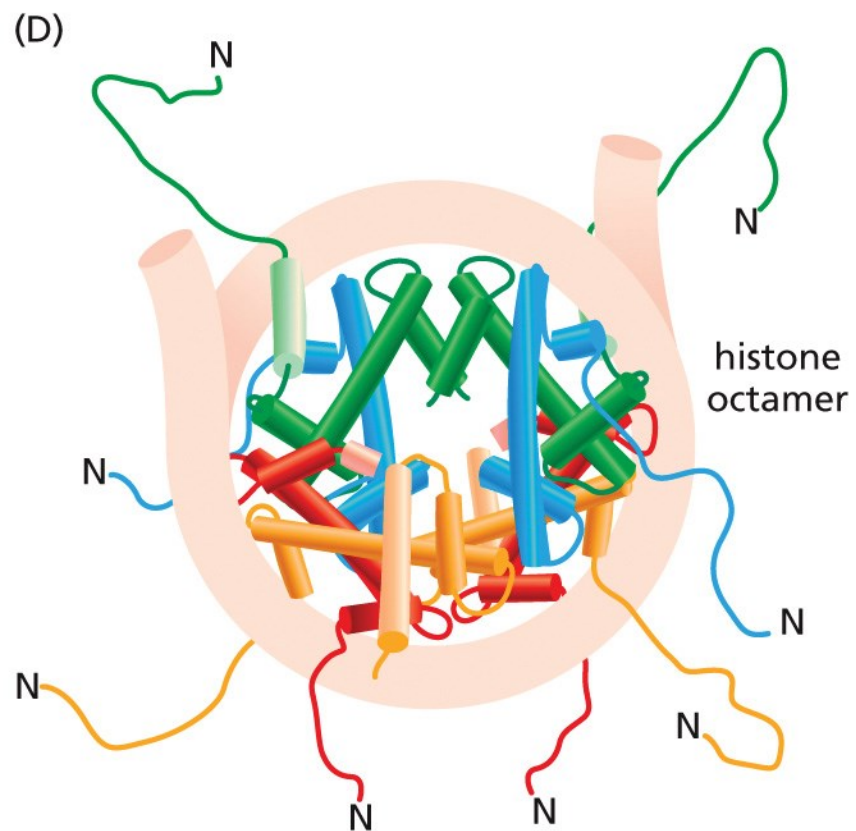
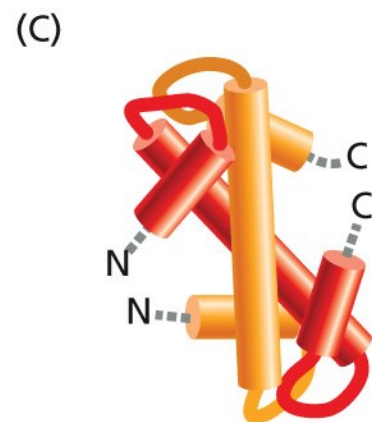
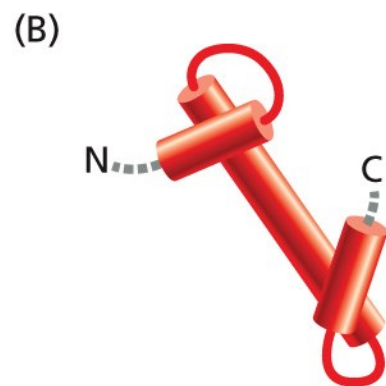
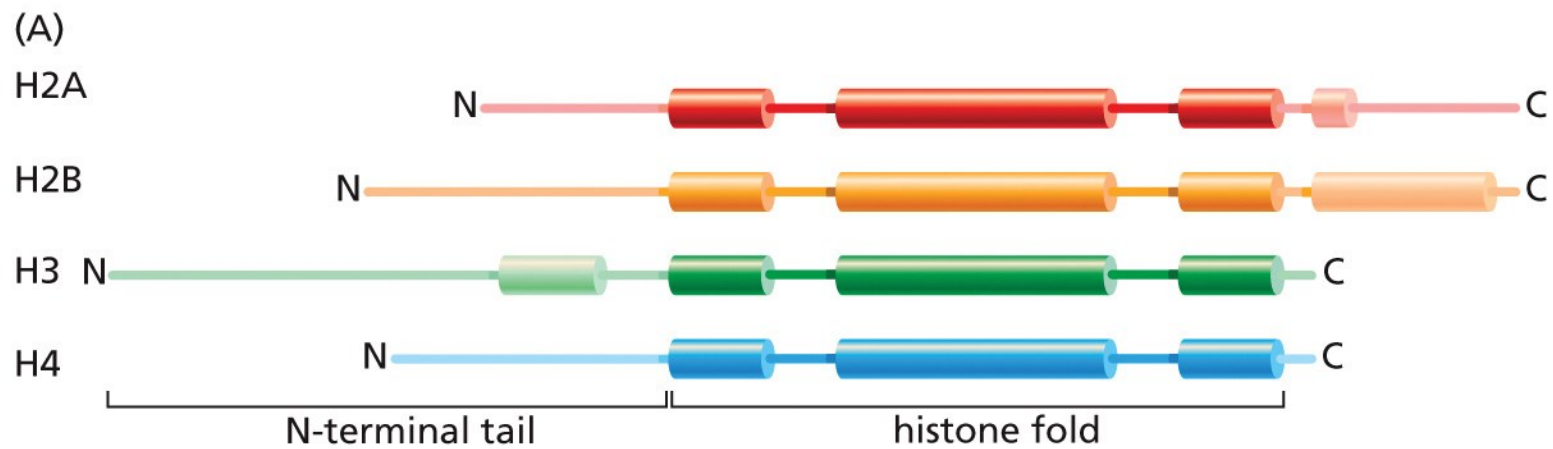
DISSOCIATION WITH HIGH CONCENTRATION OF SALT

histone octamer

147-nucleotide-pair DNA double helix

DISSOCIATION





short region of
DNA double helix



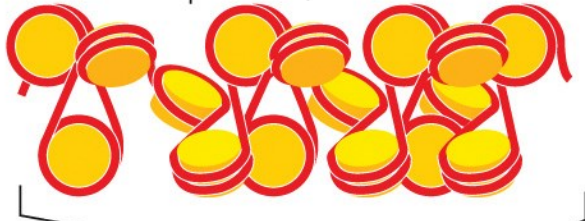
2 nm

"beads-on-a-string"
form of chromatin



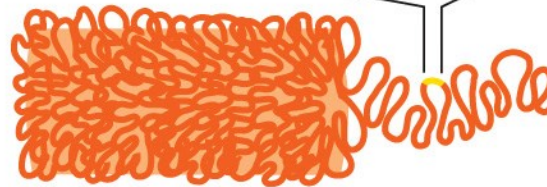
11 nm + histone H1

chromatin fiber
of associated
nucleosomes



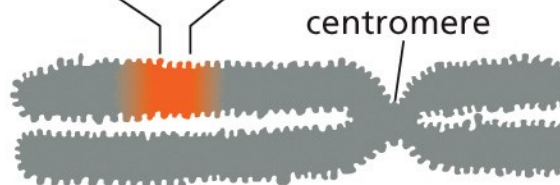
30 nm

chromatin fiber
folded into loops



700 nm

entire
mitotic
chromosome



centromere

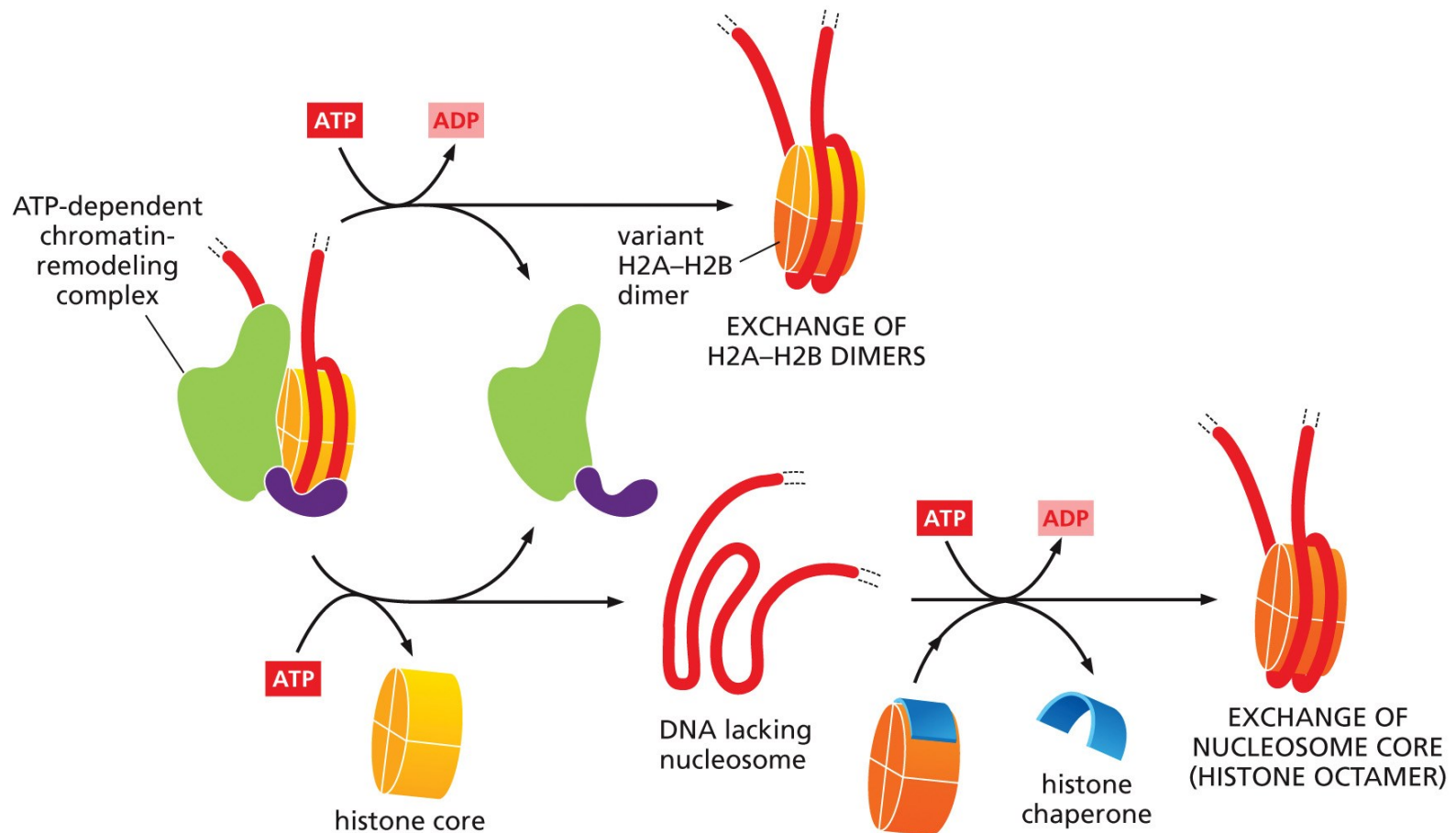
1400 nm

Exact structure
unknown

NET RESULT: EACH DNA MOLECULE HAS BEEN
PACKAGED INTO A MITOTIC CHROMOSOME THAT
IS 10,000-FOLD SHORTER THAN ITS FULLY
EXTENDED LENGTH

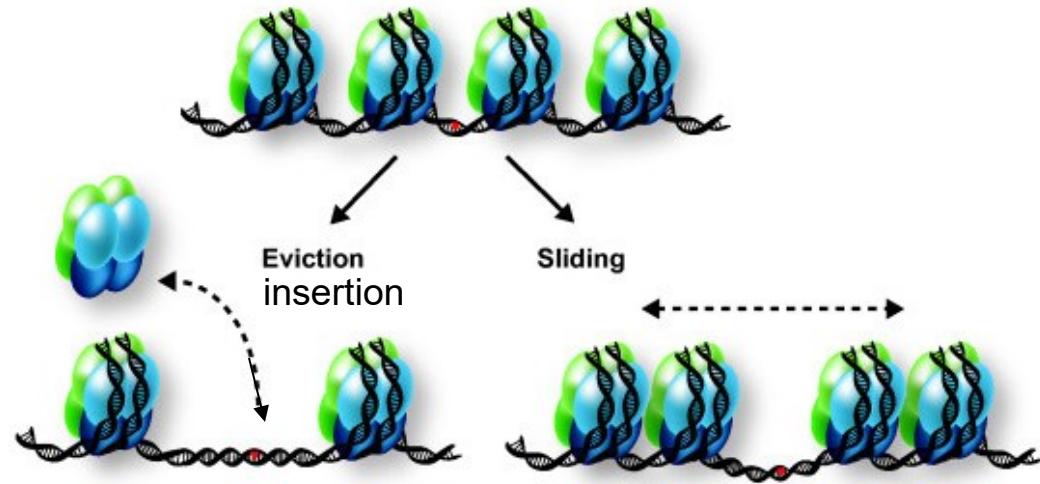
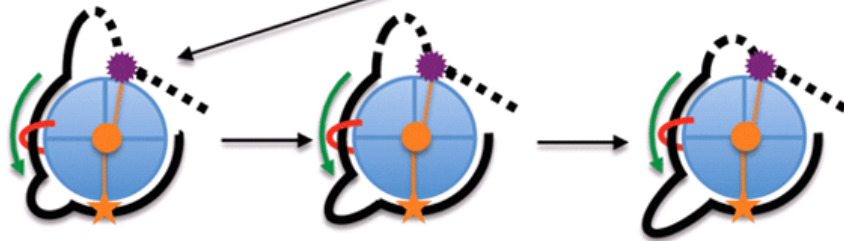
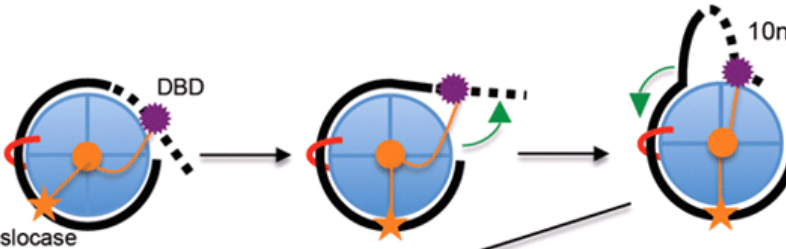
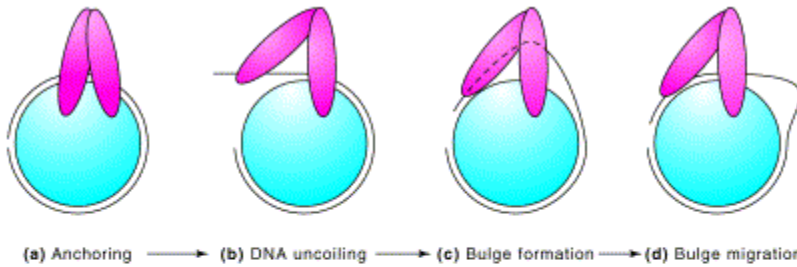
THE REGULATION OF CHROMATIN STRUCTURE:

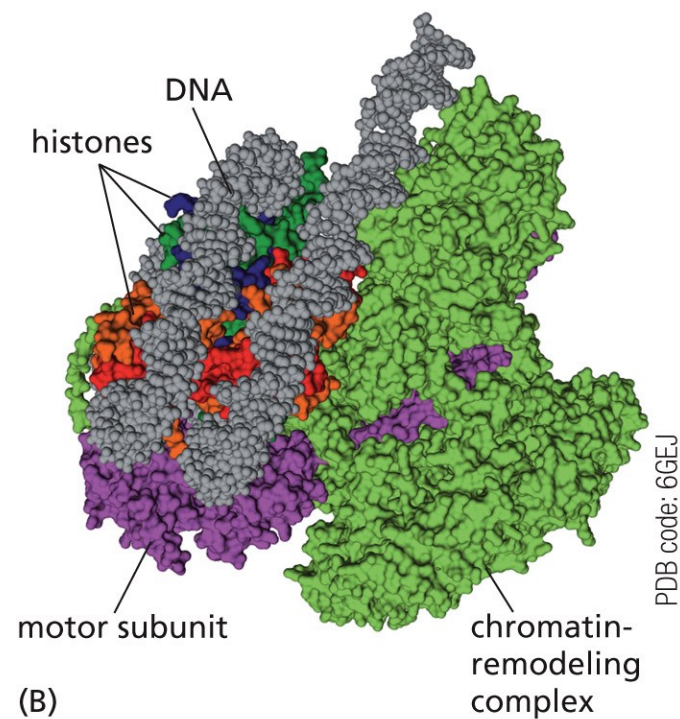
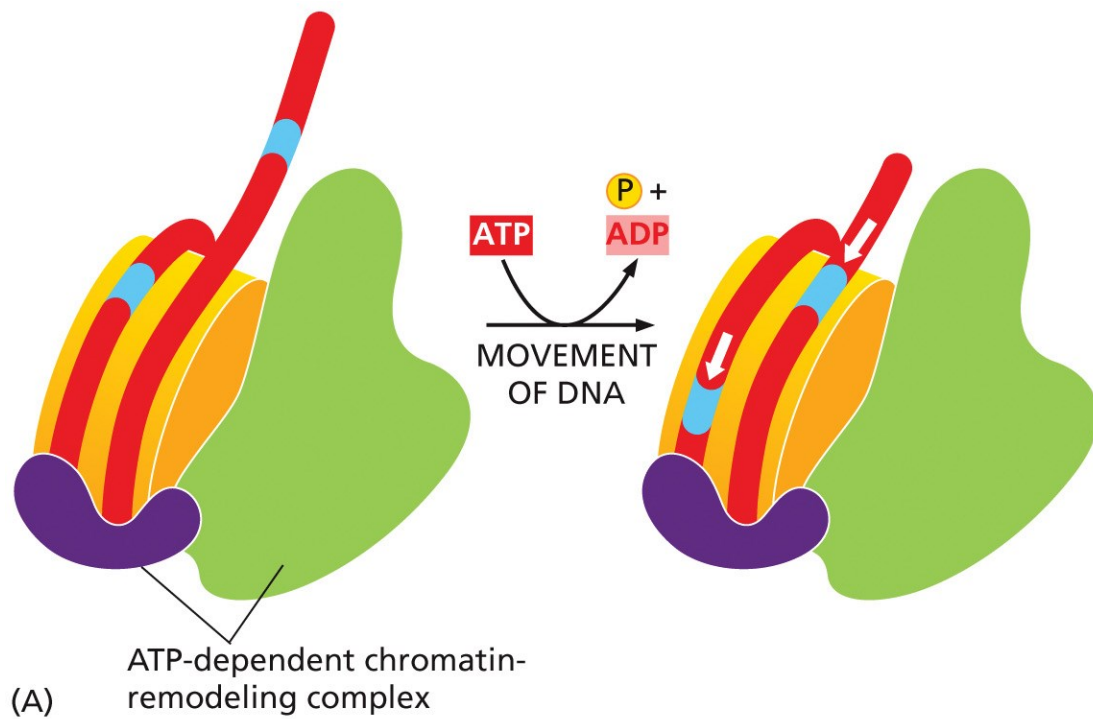
(1) ATP-DEPENDENT REMODELERS AND HISTONE CHAPERONES



ATP-dependent chromatin remodelling

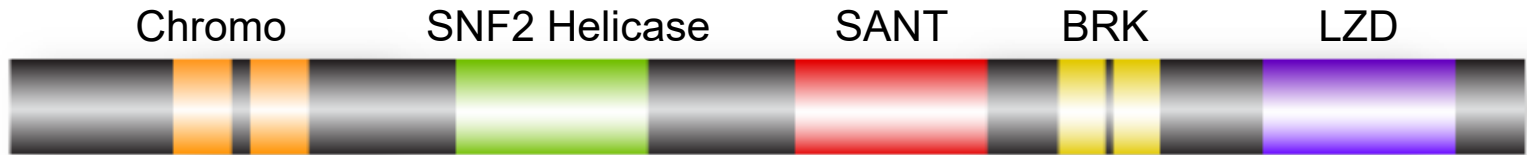
ATP-dependent remodelers





Example of an ATP-dependent chromatin remodeling factor:

Chromodomain Helicase DNA binding protein 7 (Chd7)



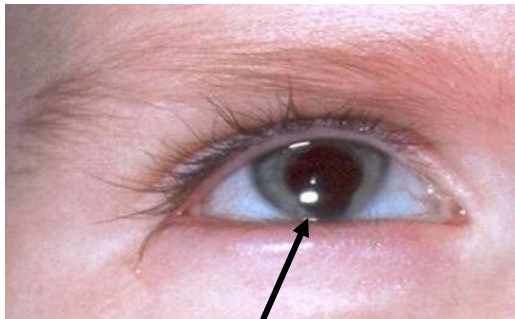
- Chd7 protein: 2986aa, ~ 330kDa
- contains SANT domain to bind histones
- contains chromodomains to bind H3K4me
- contains SNF2-like ATP-dependent helicase domain to move nucleosomes

CHD7 mutations cause CHARGE syndrome

CHARGE syndrome:

(Coloboma, Hearing loss, Atrichiasis of eyelids, Atrichiasis of trachea and esophagus, Retardation of growth, Genital anomalies, Ear anomalies)

Prevalence: 1 in 8,500-10,000



Coloboma



THE REGULATION OF CHROMATIN STRUCTURE

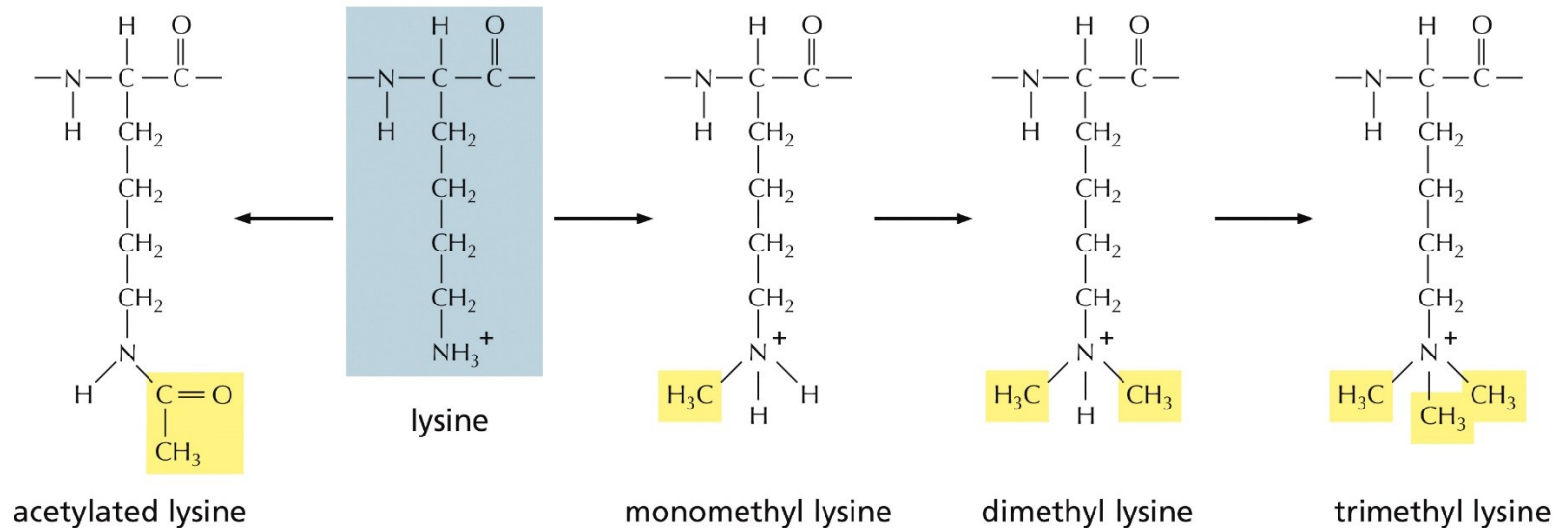
- The Core Histones Are Covalently Modified at Many Different Sites

THE REGULATION OF CHROMATIN STRUCTURE

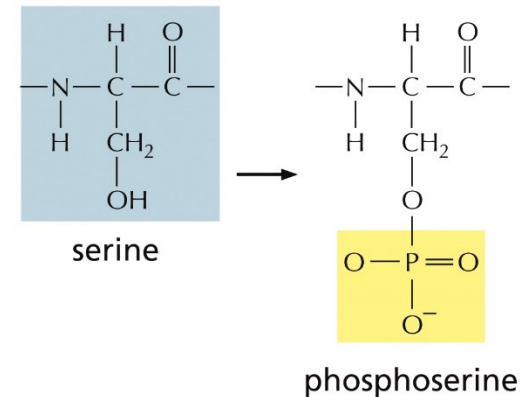
- The Covalent Modifications produce a “Histone Code” That Helps to Determine Biological Function
- A Complex of Code-reader and Code-writer Proteins Can Spread Specific Chromatin Modifications for Long Distances Along a Chromosome

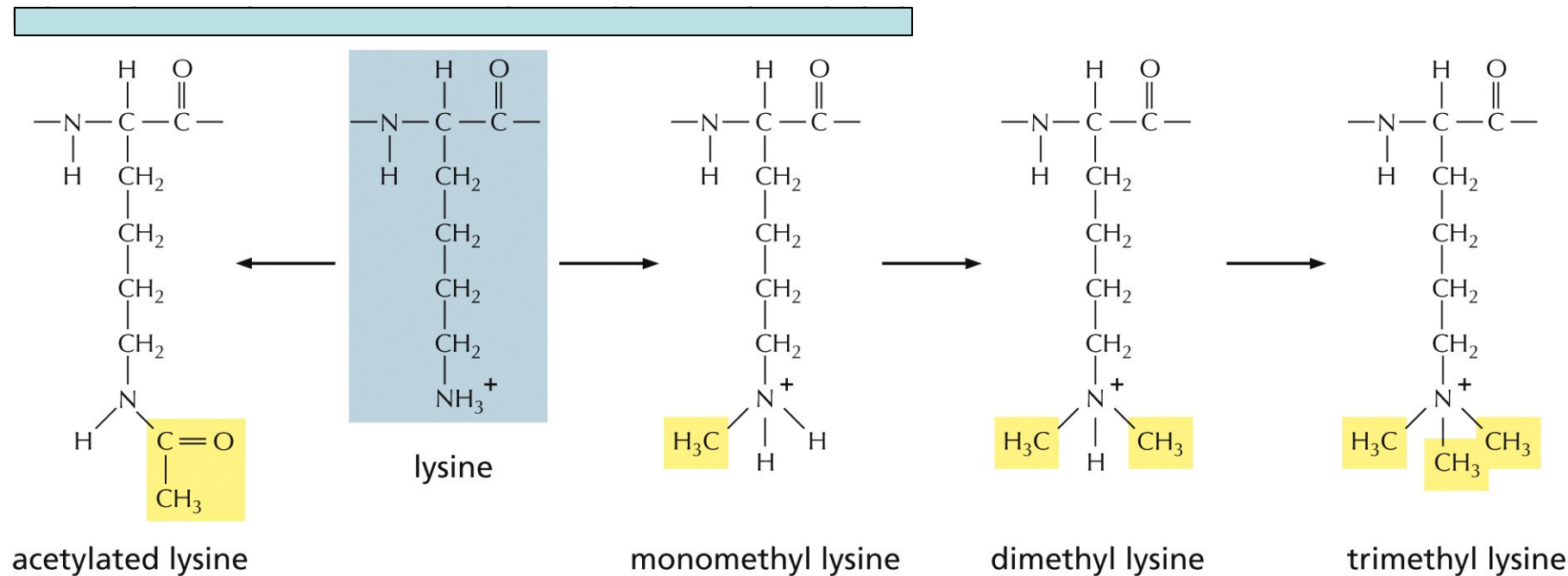
HISTONE MODIFICATIONS

(A) LYSINE ACETYLATION AND METHYLATION ARE COMPETING REACTIONS



(B) SERINE PHOSPHORYLATION





Article

Acetyl-methyllysine marks chromatin at active transcription start sites

Methylation and acetylation can occur on same histone lysine

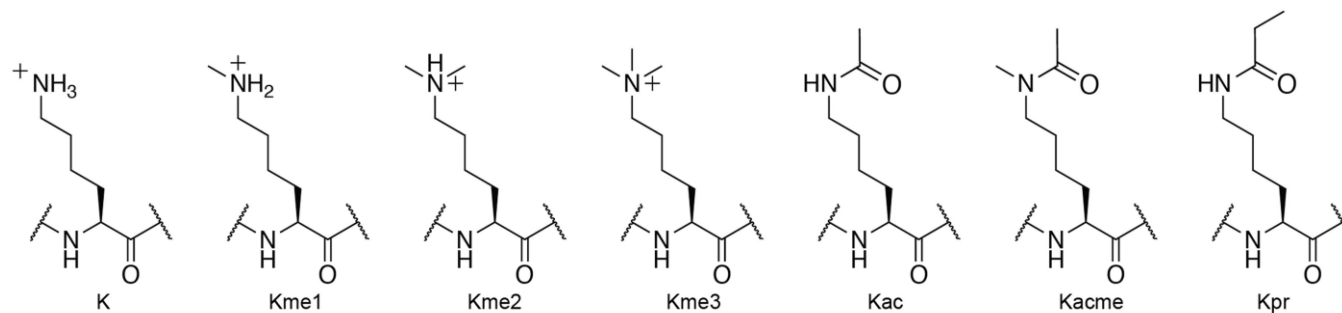
<https://doi.org/10.1038/s41586-023-06565-9>

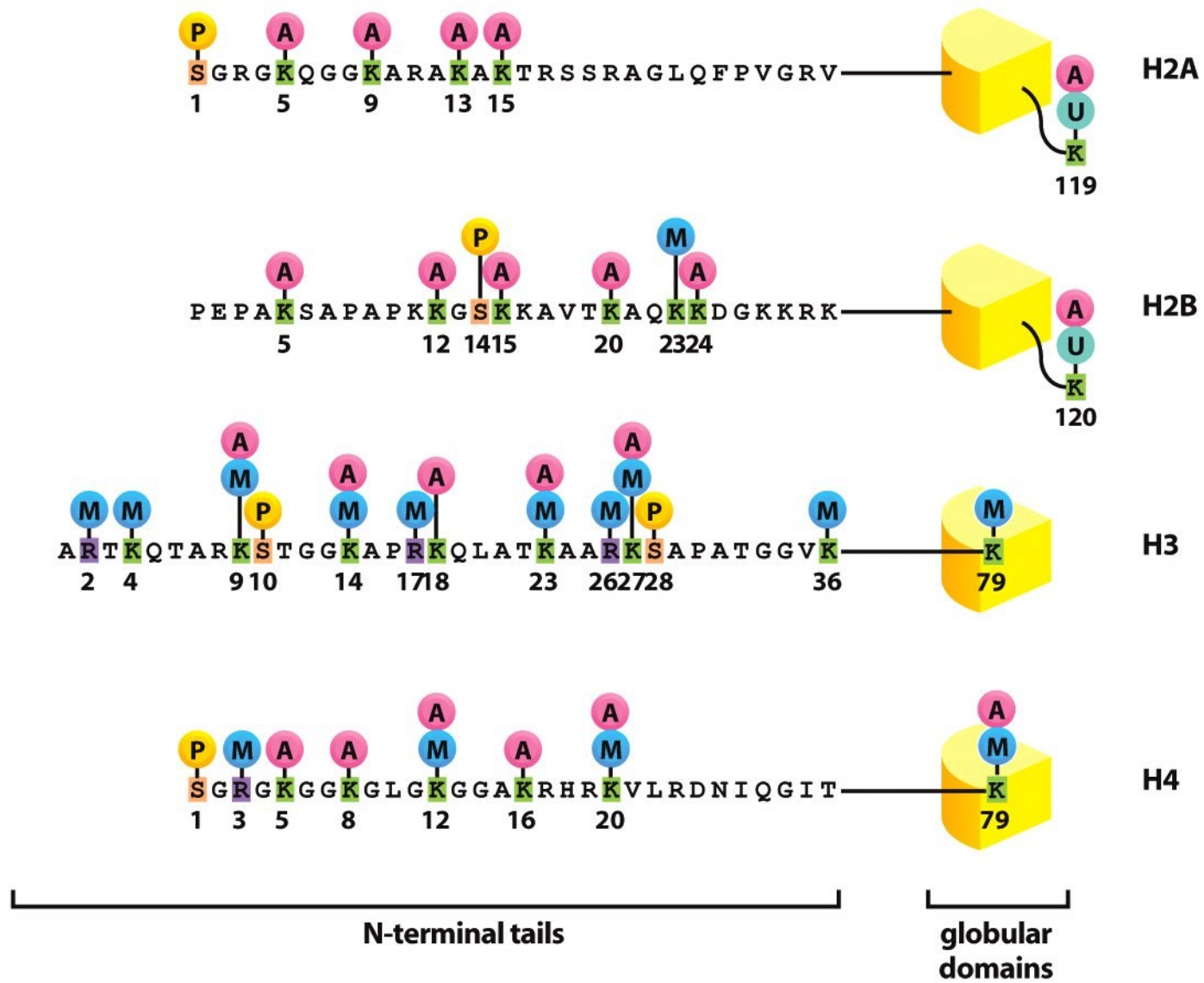
Received: 19 August 2021

Accepted: 23 August 2023

Published online: 20 September 2023

William J. Lu-Cutliff^{1,2,3}, Leah J. Connor^{1,2,3}, Yixuan Xie², Babatunde E. Ekundayo¹, Brendan T. Rose^{1,2}, Martin Machyna^{1,2}, Andreas P. Pintado-Urbanc^{1,2}, Joshua T. Zimmer^{1,2}, Isaac W. Vock^{1,2}, Natarajan V. Bhanu², Megan C. King⁴, Benjamin A. Garcia², Franziska Bleichert¹ & Matthew D. Simon^{1,2,3}





modification state

"meaning"

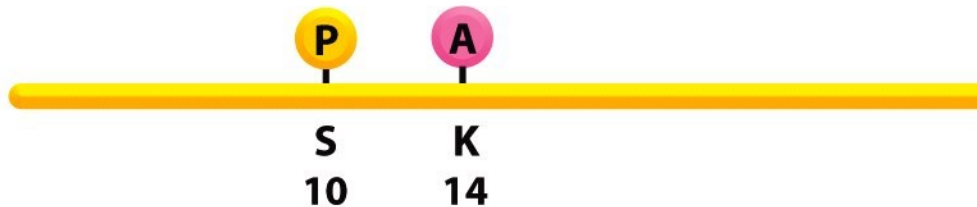


**heterochromatin formation,
gene silencing**
(H3K9me3)



gene expression

H3K4me3 = active promoter



gene expression



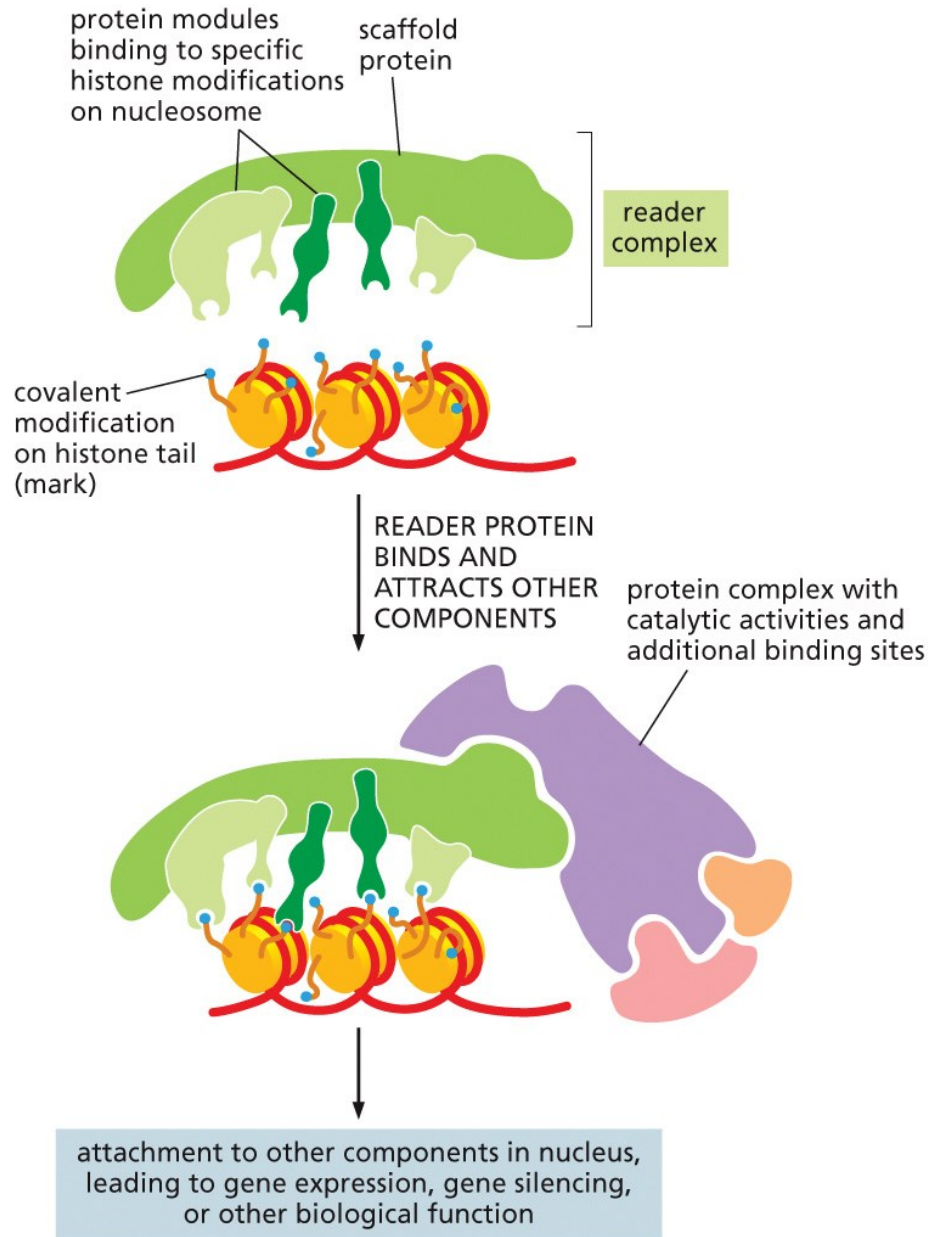
**silencing of Hox genes,
X chromosome inactivation**

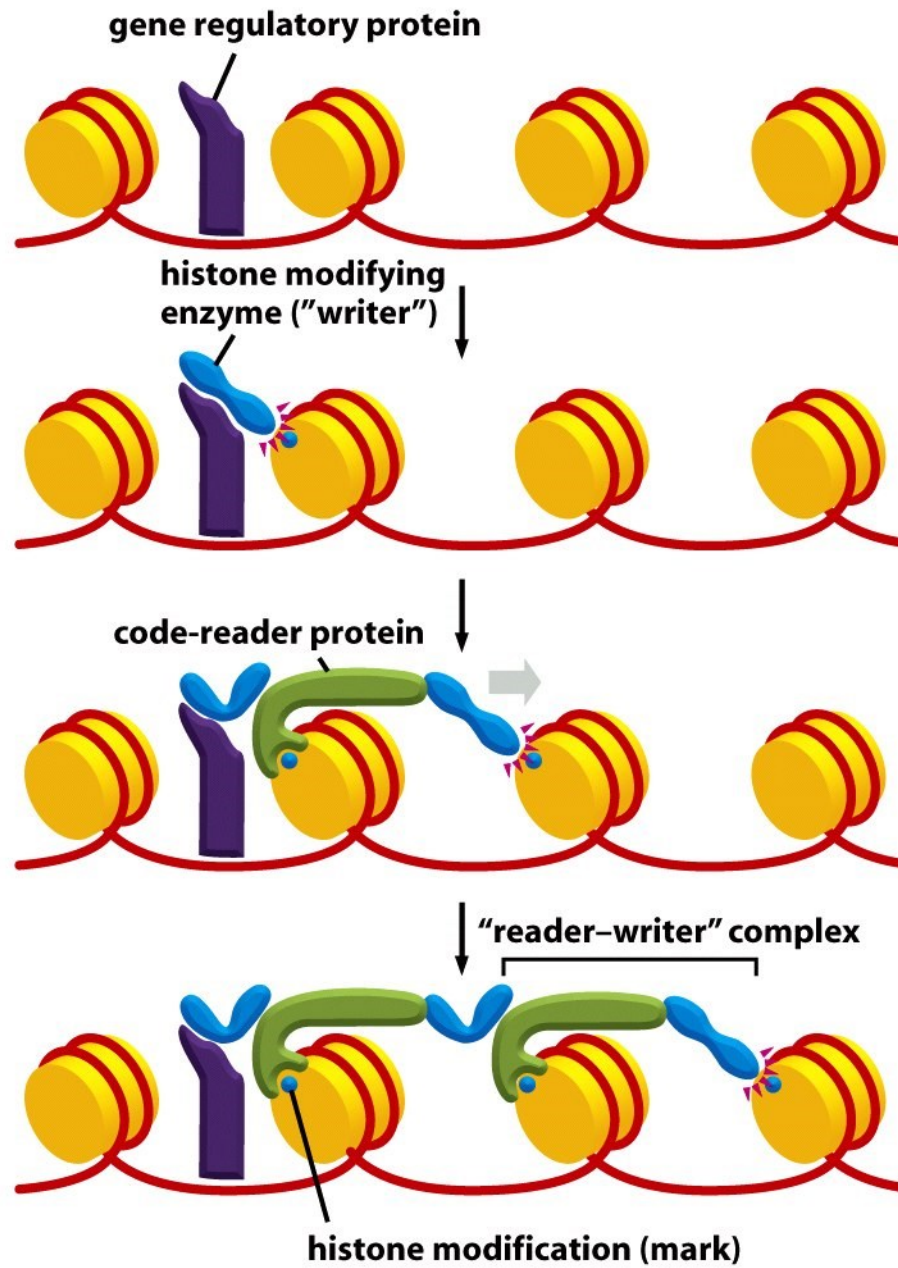
TABLE 4–2 Some Effects of Histone Modifications

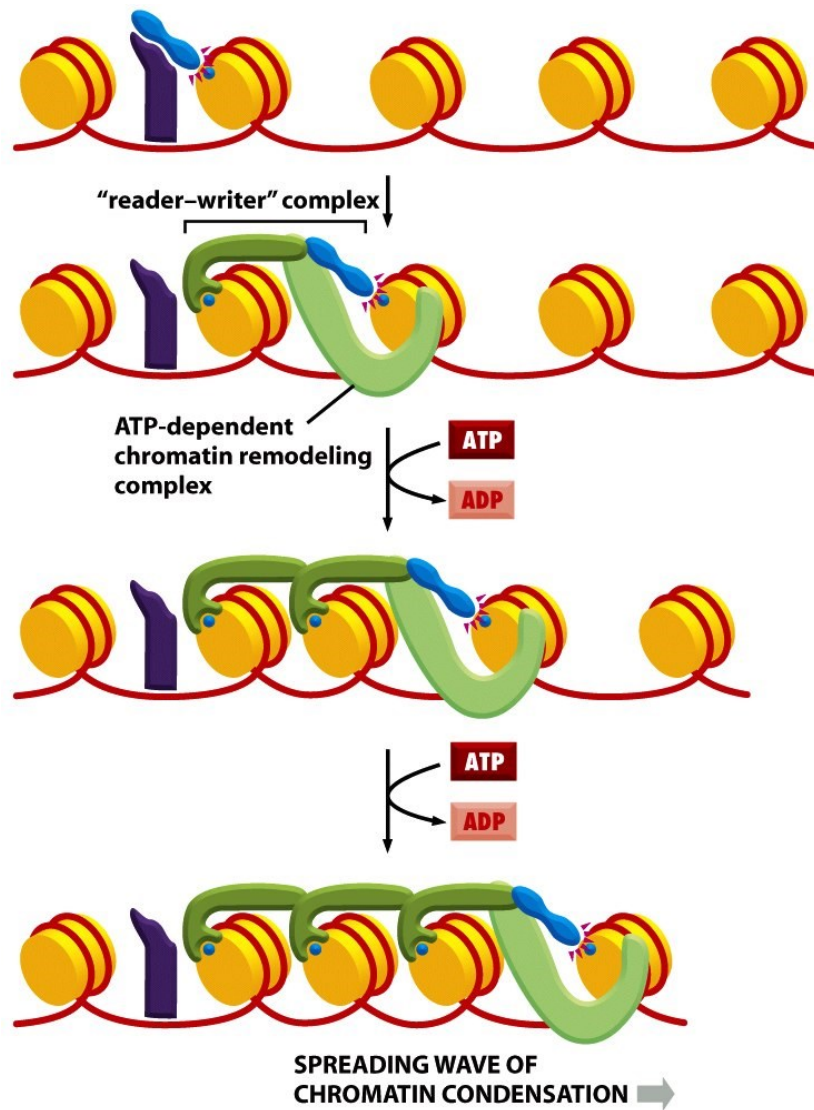
Histone modification	Association with chromatin type	Gene expression	Abundance (%)
H4K4me3	Highly accessible, open chromatin	ON	1
H3K9ac	Highly accessible, open chromatin	ON	1
H3K9me3	Heterochromatin (either constitutive or facultative)	OFF	25
H3K27me3	Facultative heterochromatin	OFF	13

In general, sets of modifications act in combination, but only a small number of their meanings are clear. For how the two different kinds of heterochromatin listed here form, see Figure 4–40.

HISTONE CODE: READERS AND WRITERS

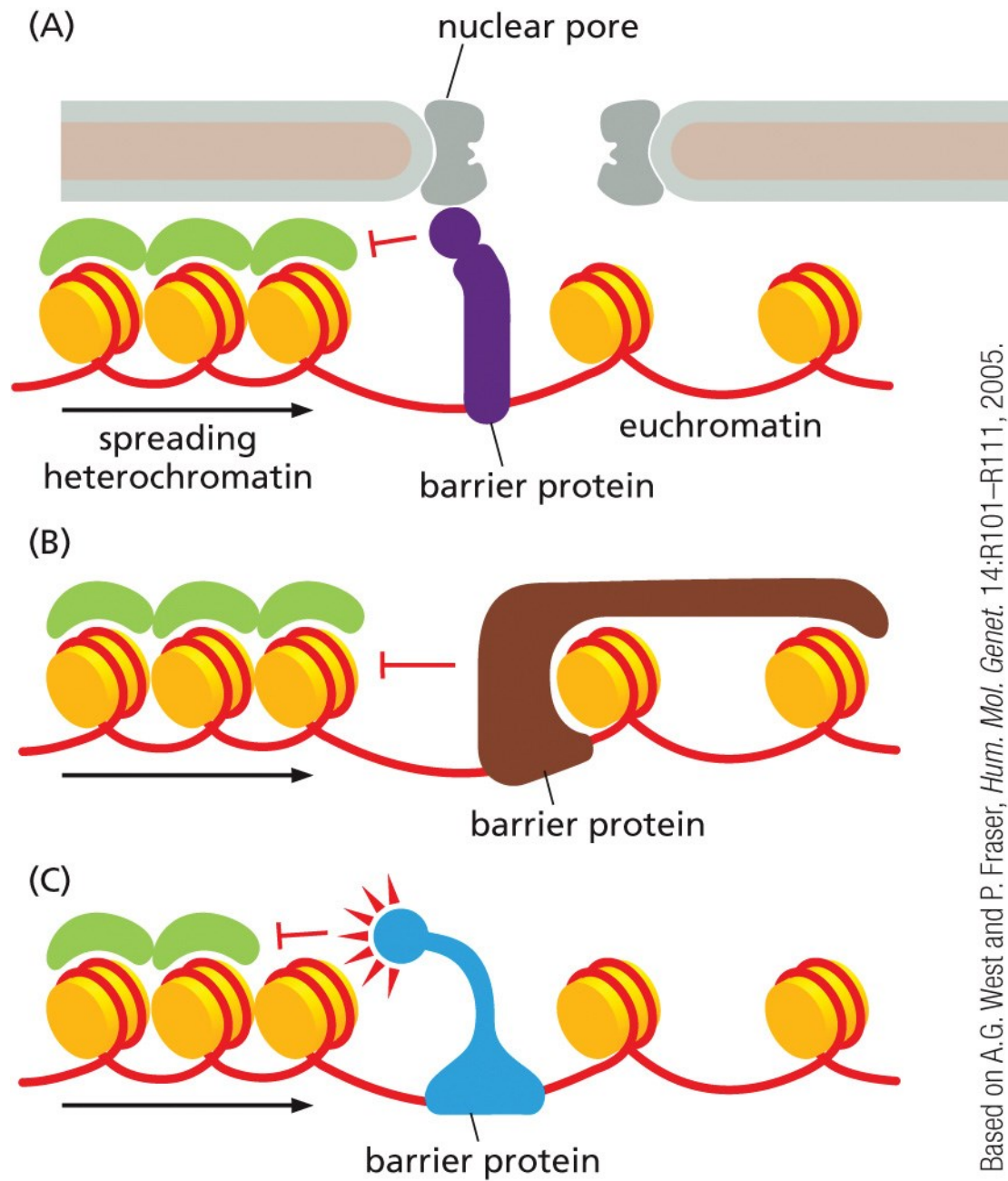






THE REGULATION OF CHROMATIN STRUCTURE

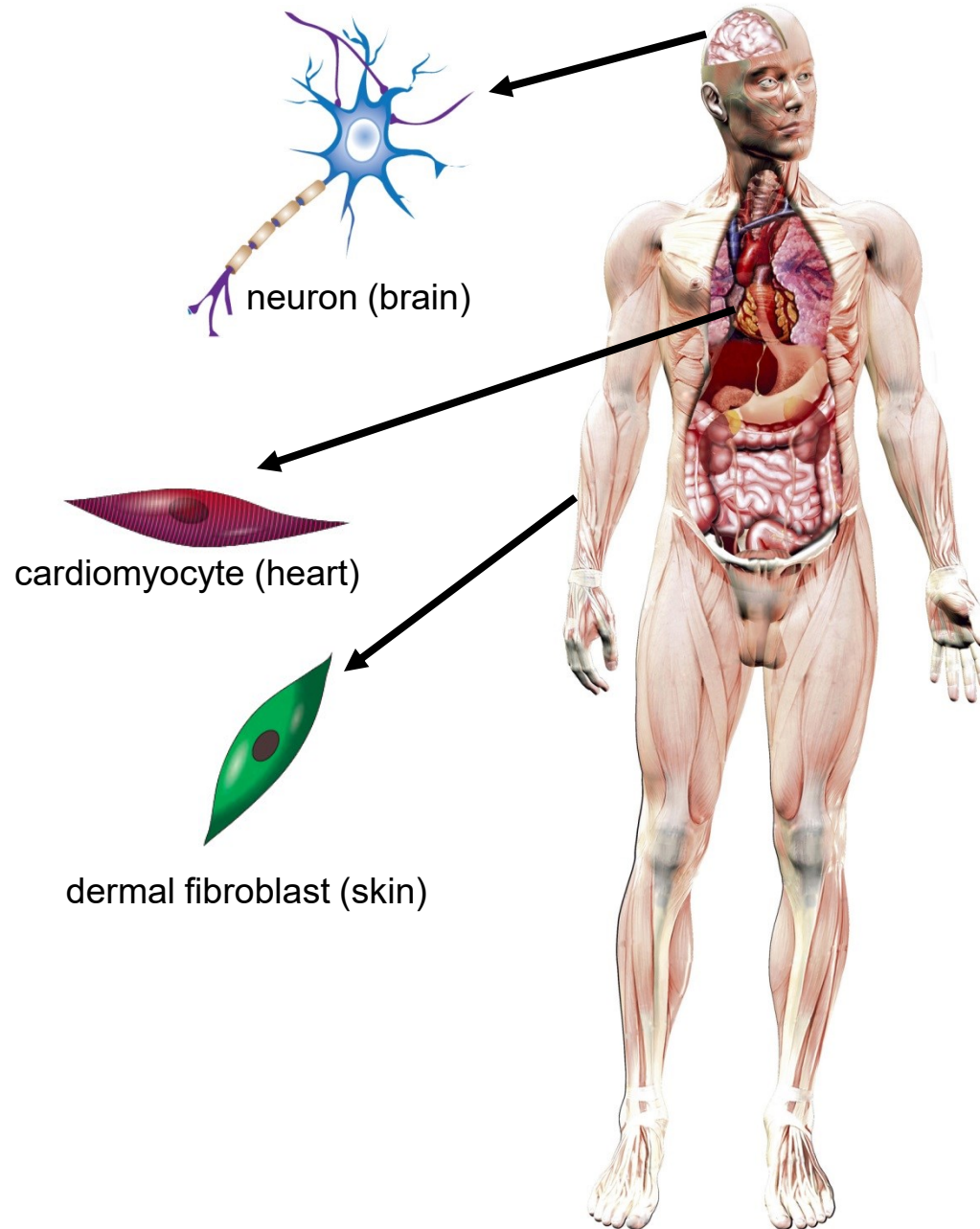
- Barrier DNA Sequences Block the Spread of Reader–Writer Complexes and thereby Separate Neighboring Chromatin Domains



Based on A.G. West and P. Fraser, *Hum. Mol. Genet.* 14:R101–R111, 2005.

What is this complex organisation of chromatin good for?

Humans contain 300 different cell types with the same DNA



How is chromatin marked in different cells?

ENCODE: Encyclopaedia of DNA Elements

International consortium to map functional elements/regions in human genome

- Analyses of 147 different human cell types and tissues
- Transcribed genes (RNA-seq)
- **Histone modifications (ChIP-seq)**
- Transcription factor binding sites (ChIP-seq)
- Accessible regions (DNase-seq)
- Transcription factor binding sites (DNase-seq)
- Chromatin 3D interactions (5C)
- Map of functional elements is a “signature” of the identity of a cell

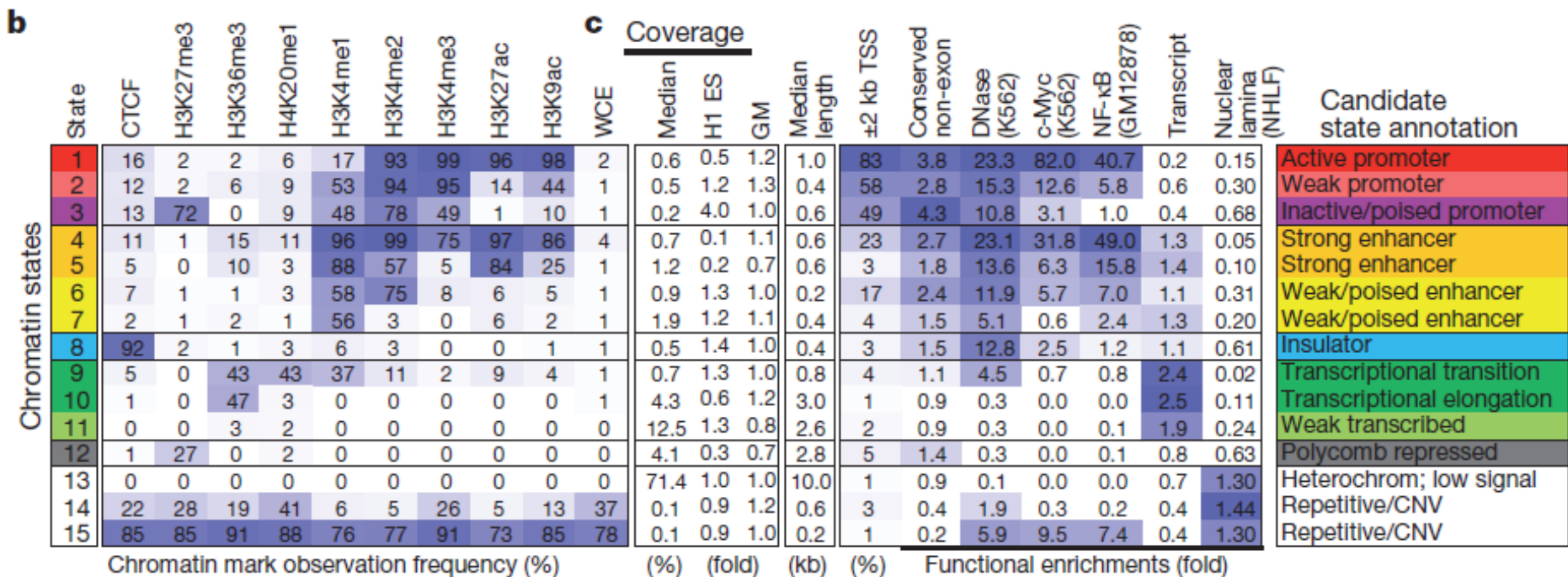


6 Sept. 2012

Mapping and analysis of chromatin state dynamics in nine human cell types

Jason Ernst^{1,2}, Pouya Kheradpour^{1,2}, Tarjei S. Mikkelsen¹, Noam Shores¹, Lucas D. Ward^{1,2}, Charles B. Epstein¹, Xiaolan Zhang¹, Li Wang¹, Robbyn Issner¹, Michael Coyne¹, Manching Ku^{1,3,4}, Timothy Durham¹, Manolis Kellis^{1,2*} & Bradley E. Bernstein^{1,3,4*}

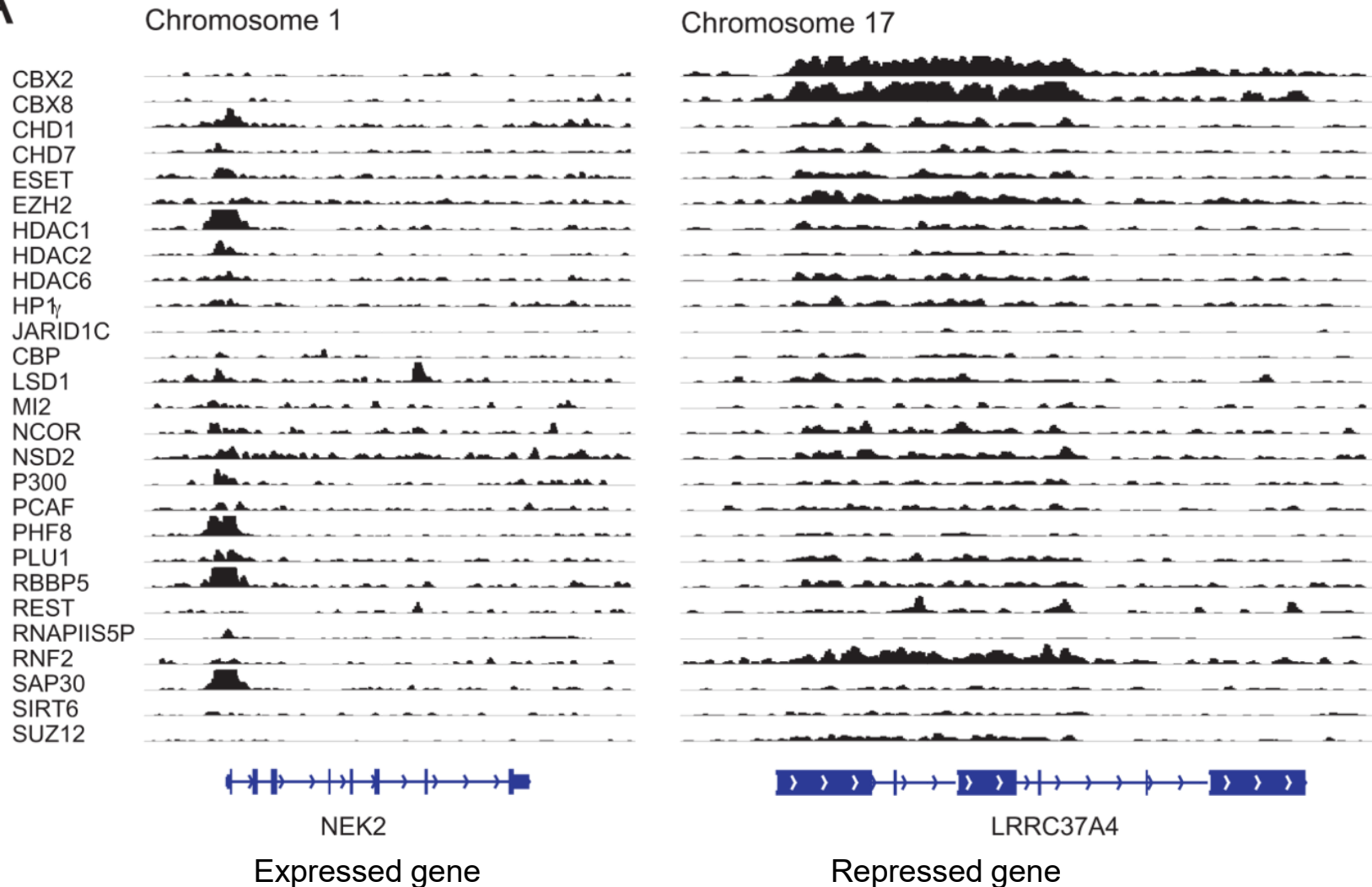
Can we assign function to parts of our genome by the histone modifications that are present?



Combinatorial Patterning of Chromatin Regulators Uncovered by Genome-wide Location Analysis in Human Cells

Oren Ram,^{1,2,3,4,7} Alon Goren,^{1,2,3,4,7} Ido Amit,^{1,2} Noam Shores,¹ Nir Yosef,^{1,2} Jason Ernst,^{1,5} Manolis Kellis,^{1,5} Melissa Gymrek,^{1,2,3,4} Robbyn Issner,¹ Michael Coyne,¹ Timothy Durham,¹ Xiaolan Zhang,¹ Julie Donaghey,¹ Charles B. Epstein,¹ Aviv Regev,^{1,2,6,*} and Bradley E. Bernstein^{1,2,3,4,*}

A



Systematic Localization of Common Disease-Associated Variation in Regulatory DNA

Matthew T. Maurano,^{1*} Richard Humbert,^{1*} Eric Rynes,^{1*} Robert E. Thurman,¹ Eric Haugen,¹ Hao Wang,¹ Alex P. Reynolds,¹ Richard Sandstrom,¹ Hongzhu Qu,^{1,2} Jennifer Brody,³ Anthony Shafer,¹ Fidencio Neri,¹ Kristen Lee,¹ Tanya Kutayavin,¹ Sandra Stehling-Sun,¹ Audra K. Johnson,¹ Theresa K. Canfield,¹ Erika Giste,¹ Morgan Diegel,¹ Daniel Bates,¹ R. Scott Hansen,⁴ Shane Neph,¹ Peter J. Sabo,¹ Shelly Heimfeld,⁵ Antony Raubitschek,⁶ Steven Ziegler,⁶ Chris Cotsapas,^{7,8} Nona Sotoodehnia,^{3,9} Ian Glass,¹⁰ Shamil R. Sunyaev,¹¹ Rajinder Kaul,⁴ John A. Stamatoyannopoulos^{1,12†}

GWAS; Genome-wide association studies

Disease-associated variants often outside coding region

Problems:

- which SNP is real?; co-segregation multiple SNPs
- in which tissue does the SNP “act”

Active regulatory regions span 2% of genome, contain 70% of disease associated SNPs

